Our Contacts:
Zhalimbetov Askhat – CEO
Baimuldinova Nazira – Project Manager

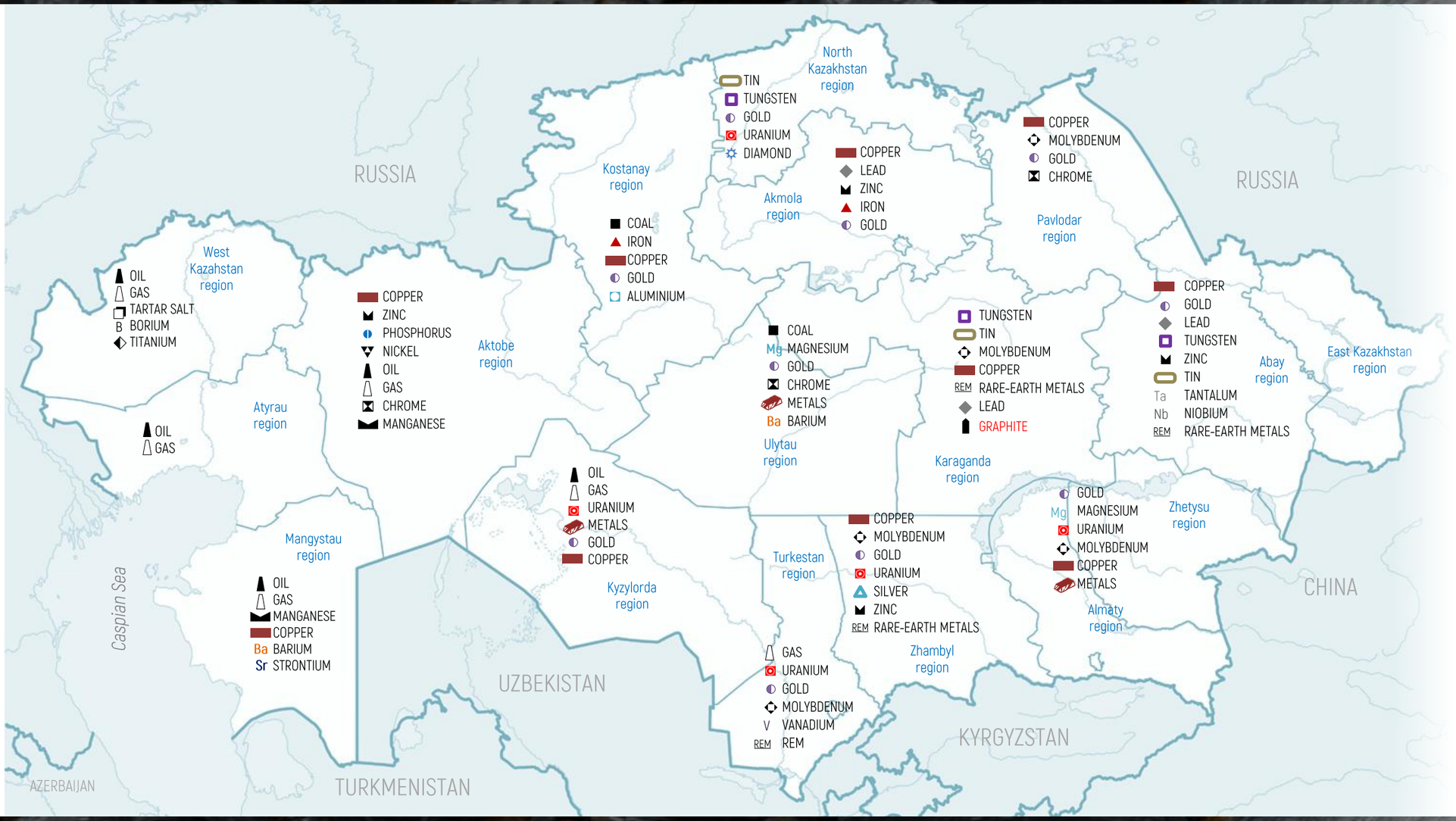
E-mail: mining.kz2021@gmail.com, holart.kz@gmail.com
Mob.: + 7 775 85 88 387

GRAPHITE

exploration in the Karaganda Region of the Republic of Kazakhstan

SUBSOIL ASSETS OF REPUBLIC OF KAZAKHSTAN

KGM KAZGEOMINER

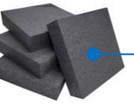


RioTinto
CNOOC
YILDIRIM HOLDING INC.
ULMUS FUND
ArcelorMittal
CITIC Group
Eni
EVRAZ
Shell
РОСНЕФТЬ
LUKOIL
SARYTOGAN GRAPHITE
CNPC
JIANGXI COPPER CORPORATION
ExxonMobil
POLYMETAL
NFC
GLENCORE
SINOPEC
Chevron
TOTAL
KAZ MINERALS
KATKO
ERG
FMG Fortescue
WorleyParsonsTWP resources & energy

GLOBAL GRAPHITE DEMAND AND MAJOR SUPPLIERS

KGM KAZGEOMINER

GRAPHITE is a unique carbon material with excellent thermal conductivity, high-temperature resistance, and outstanding electrical conductivity. It is used across various industries, from traditional to high-tech sectors:

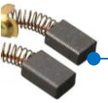


The refractory industry accounts for nearly half of global consumption due to graphite's ability to withstand extreme temperatures (bricks, furnace linings).

Batteries for electric vehicles and ESS especially for lithium-ion batteries; graphite is widely used in anodes, and its share in the battery industry's consumption is rapidly growing.

Metallurgy and foundry
electrodes and linings for steel, aluminum, and other metals.

Lubricants, brake friction materials, composites
graphite materials are used for their anti-friction properties.



Other niches
high conductivity makes graphite useful in electronics, powder metallurgy, and fire-retardant additives.

Agriculture (Fertilizers) - nutrient carrier, component of slow-release fertilizers, improvement of soil structure, enhancement of water retention capacity and efficiency of micronutrient uptake by plants

Defense Industry and Ballistics - production of heat-resistant composite materials, lubricating components for ammunition, composites and reinforced structures for body armor and protective equipment, etc.

China controls the overwhelming share of graphite mining and processing: in 2024–2025, about 77–78% of global natural graphite production was attributed to China, and nearly all processing facilities are also concentrated there

Madagascar is the second-largest producer (about 89,000 tons in 2024) and holds significant reserves

Mozambique is one of Africa's leading producers, with the large Balama project being one of the largest in the world

Brazil is a notable supplier, also ranking among the TOP-5 producers

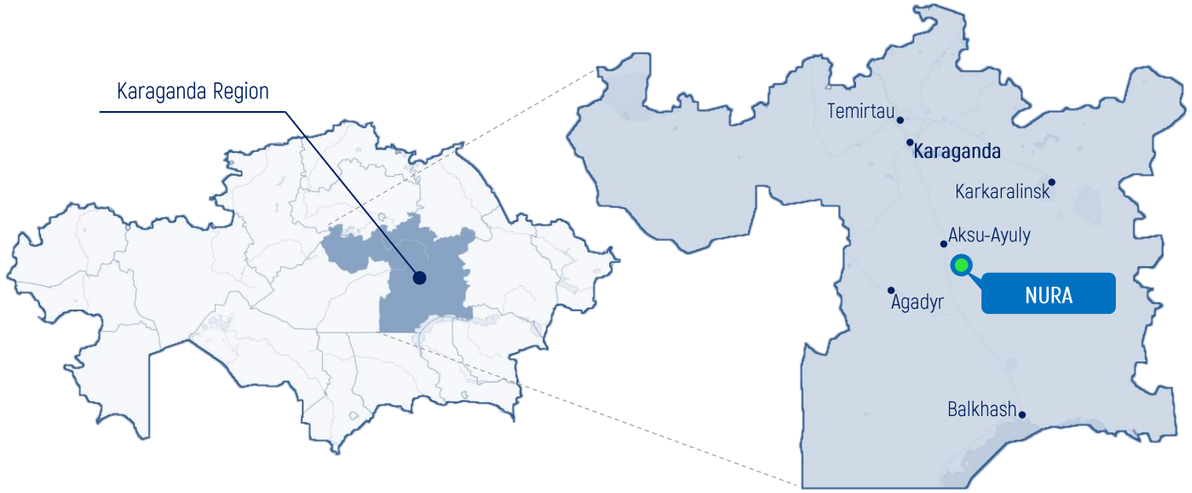
India produces several tens of thousands of tons of graphite annually

GLOBAL GRAPHITE DEMAND AND MAJOR SUPPLIERS

SITE LOCATION MAP

Location: 55 km southeast of Aksu-Ayuly settlement (center of Shet District), 12 km southeast of Nura village

Infrastructure: A national highway (R-36) runs 45 km west of the site. A paved local road (R-181, Aksu-Ayuly–Aktogay) passes 6 km north of the site. The nearest railway station is 110 km away. Electricity and water supply can be provided from local sources. The logistics allow for a transition to industrial-scale operations without capital-intensive investments.



The AIFC is not only a financial hub, but also an attractive business platform with international standards, legal protection, tax incentives, a flexible currency environment, and access to capital for companies, investors, and professionals

Existing facilities:

- The nearest mining enterprise is the Almalinsky mine for copper and gold-bearing ores (25 km north);
- The Sarytogambay site (Sarytogan Graphite Limited) is located 10 km northeast of the Nura site.

Stage: Exploration and evaluation works (2025–2031)

License Area:
13.67 km²

License:
No. 3688-EL dated Oct. 4, 2025

Subsoil User:
KazGeoMiner LLP

Mineral resources: P3 category - 71.2 mln.tonnes of ore, containing approximately 200 thous. tonnes of graphite, with an average grade of 28,8%

GRAPHITE EXPLORATION IN THE KARAGANDA REGION OF THE REPUBLIC OF KAZAKHSTAN

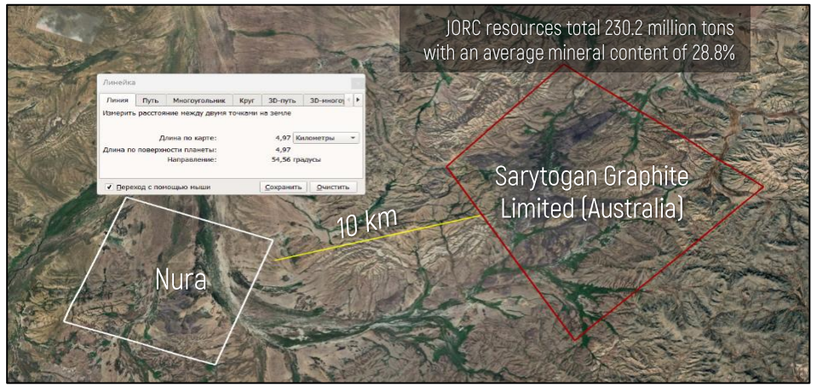
Geological Information: The project is aimed at identifying and assessing natural graphite reserves in a geologically prospective area, which is a continuation of the existing **Sarytogambay graphite district (Sarytogan Graphite Limited).**

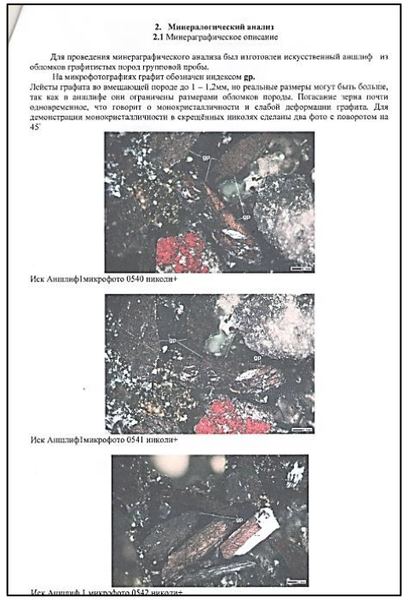
The Nura site is the southwestern extension of the Sarytogambay deposit, where graphite has already been confirmed at an industrial scale. Graphite is associated with carbonaceous shale and coal-bearing strata of the Middle Devonian. The rocks have undergone regional and contact metamorphism, favorable for graphitization.

Samples with graphite content of up to 99.7% have been recorded (based on historical data). The geological model confirms a high likelihood of discovering extensive graphite-bearing bodies.

In 2024, a test report from the K. I. Satpayev Institute indicated that the examined sample has a high carbon content: graphite in various morphologies (crystalline and cryptocrystalline) **accounts for approximately 47% of the total mass.**

Associated minerals (carbonates, kaolinite, quartz) play a minor role. Zircon is present only as isolated inclusions.





2. Минералогический анализ
2.1 Минераграфическое описание

Для проведения минераграфического анализа был изготовлен искусственный аншлиф из обломков графитистых пород групповой пробы.

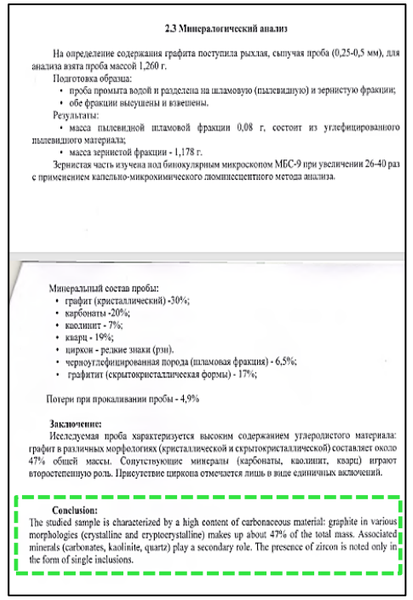
На микрофотографиях графит обозначен индексом **gp.**

Лейсты графита во вмещающей породе до 1 – 1,2мм, но реальные размеры могут быть больше, так как в аншлифе они ограничены размерами обломков породы. Погасание зерна почти одновременное, что говорит о монокристалличности и слабой деформации графита. Для демонстрации монокристаллічности в скрещенных николях сделаны два фото с поворотом на 45°

Иск Аншлиф1микрофото 0540 николи+

Иск Аншлиф1микрофото 0541 николи+

Иск Аншлиф 1 микрофото 0542 николи+



2.3 Минералогический анализ

На определение содержания графита поступила рыхлая, сыпучая проба (0,25-0,5 мм), для анализа взята проба массой 1,260 г.

Подготовка образца:

- проба промыта водой и разделена на шламовую (пылевидную) и зернистую фракции;
- обе фракции высушены и взвешены.

Результаты:

- масса пылевидной шламовой фракции 0,08 г, состоит из углефицированного пылевидного материала;
- масса зернистой фракции - 1,178 г.

Зернистая часть изучена под бинокулярным микроскопом МБС-9 при увеличении 26-40 раз с применением капельно-микрохимического люминесцентного метода анализа.

Минеральный состав пробы:

- графит (кристаллический) <50%;
- карбонаты - 20%;
- каолинит - 7%;
- кварц - 19%;
- циркон - редкие знаки (рзн).
- черноуглефицированная порода (шламовая фракция) - 6,5%;
- графитит (скрытокристаллическая формы) - 17%;

Потери при прокаливании пробы - 4,9%

Заключение:

Исследуемая проба характеризуется высоким содержанием углеродистого материала: графит в различных морфологиях (кристаллической и скрытокристаллической) составляет около 47% общей массы. Сопутствующие минералы (карбонаты, каолинит, кварц) играют второстепенную роль. Присутствие циркона отмечается лишь в виде единичных включений.

Conclusion:

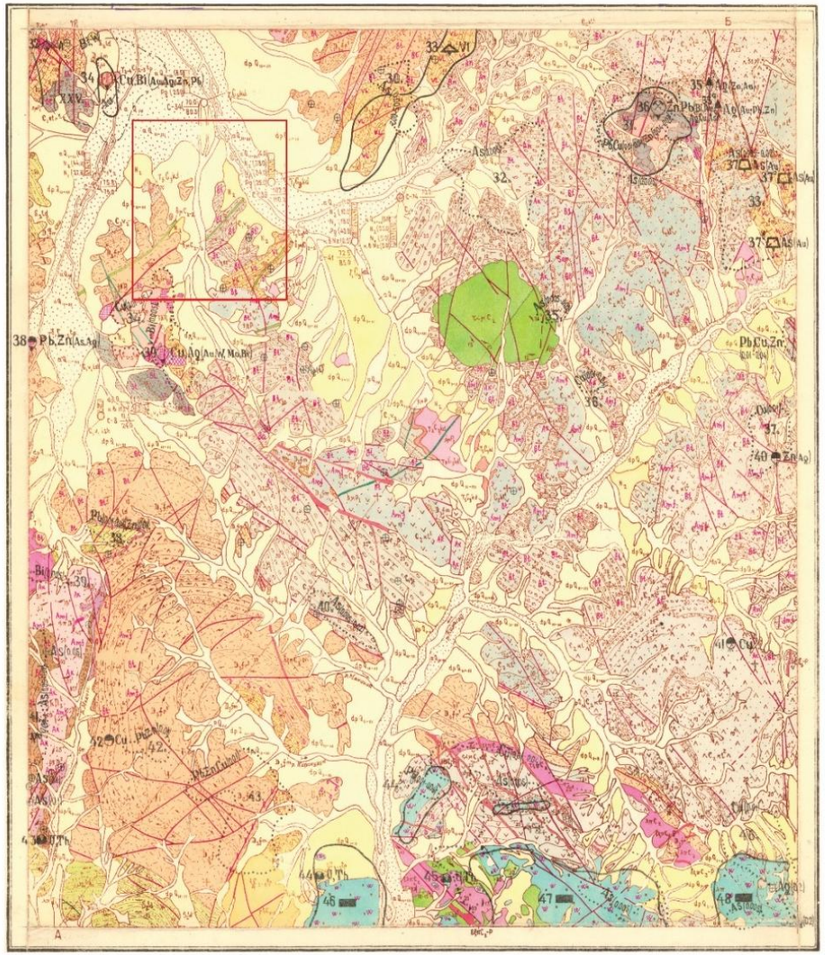
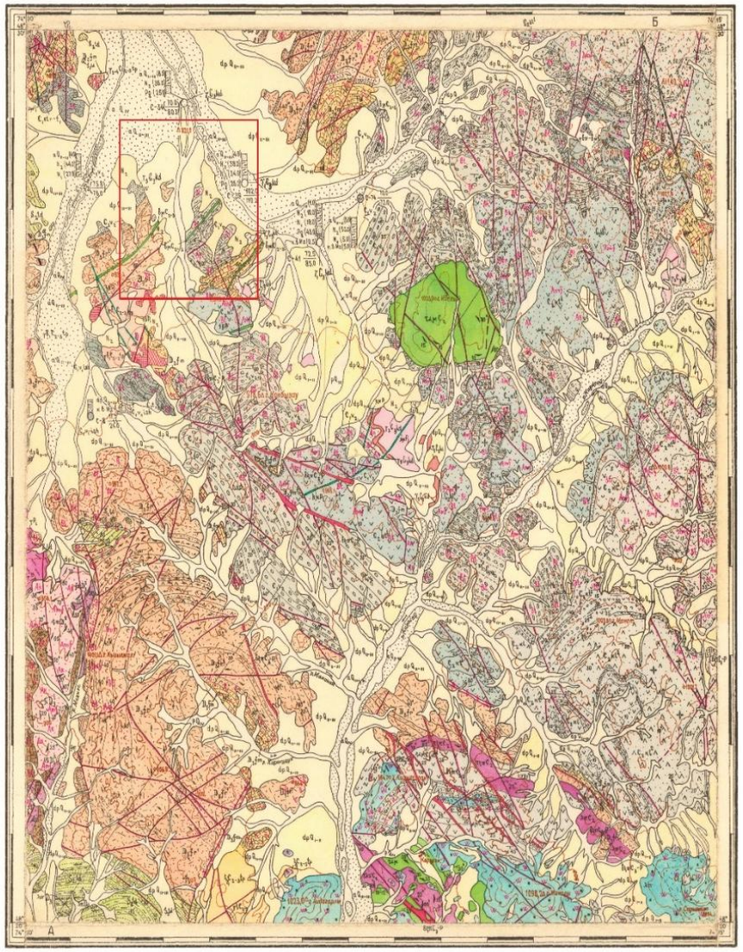
The studied sample is characterized by a high content of carbonaceous material: graphite in various morphologies (crystalline and cryptocrystalline) makes up about 47% of the total mass. Associated minerals (carbonates, kaolinite, quartz) play a secondary role. The presence of zircon is noted only in the form of single inclusions.

GEOLOGICAL STUDY OF THE SITE

1948-49 years	K. Kulikov and V. Ya. Lun mapped the northern part of sheet M-43-XXXII at a scale of 1:500,000. These researchers were the first to identify the development of coal-bearing effusive rocks.
1951-52 years	Exploration work at the Aleksandrovsk copper deposit was conducted by M. I. Aleksandrova.
1950-1967 years	Agadyr Geophysical Expedition. The entire area was surveyed using a metalometry survey at a scale of 1:50,000, partially with a ground magnetic survey at the same scale, and a gravimetric survey. In addition, the entire sheet area was included in the territory covered by the Agadyr GFE Aerial Survey Party at a scale of 1 : 25 000.
1964 years	To study rare-metal mineralization, the Stepnaya Expedition drilled several exploration wells along two profiles following the recommendation of V. A. Zlobin.
1983-84 years	The Kurpetai Party of the Central Geological Exploration Expedition (KazQuartzGemstones) conducted exploration for turquoise in the area of tourmalinization. Trenches and pits exposed layers of black carbonaceous graphitized shales. In 1984, L. M. Karimov collected **5 samples containing graphite, with mineralogical analysis indicating up to 99.7% graphite content.**
2019-2022 years	LLP "Ushtogan" (Sarytogan Graphite Limited (Australia) conducted exploration of graphite, rare, rare-earth, and precious metals in the Sarytogambay area. Detailed exploration and geological-economic assessment of graphite reserves were carried out to determine their industrial significance and evaluate the feasibility of their extraction and beneficiation at the present time.

MINERAL AND GEOLOGICAL MAPS

Mineral Map



Geological Map